

The Analysis of Successful Movies

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Team Movie: Chunyi & Nora

Why Movies & Data Cleaning

- Mainstream form of entertainment
 - Familiar topic to the audience
 - We both become nearsighted because we watch too many movies
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- Remove observations with N/A
 - Capitalized the first letter in each word
 - Remove extreme outliers (in some cases)

Dataset: Hollywood Movies

- Movies that went to theaters from 2007-2013

	Movie <chr>	LeadStudio <chr>	RottenTomatoes <int>	AudienceScore <int>
1	Spider-Man 3	Sony	61	54
2	Shrek the Third	Paramount	42	57
3	Transformers	Paramount	57	89
4	Pirates of the Caribbean: At World's End	Disney	45	74
5	Harry Potter and the Order of the Phoenix	Warner Bros	78	82

Story <chr>	Genre <chr>	TheatersOpenWeek <int>	OpeningWeekend <dbl>	BOAvgOpenWeekend <int>	DomesticGross <dbl>
Metamorphosis	Action	4252	151.100	35540	336.530
Quest	Animation	4122	121.600	29507	322.720
Monster Force	Action	4011	70.500	17577	319.250
Rescue	Action	4362	114.700	26302	309.420
Quest	Adventure	4285	77.100	17998	292.000

ForeignGross <dbl>	WorldGross <dbl>	Budget <dbl>	Profitability <dbl>	OpenProfit <dbl>	Year <int>
554.340	890.870	258.0	345.30	58.57	2007
476.240	798.960	160.0	499.35	76.00	2007
390.460	709.710	150.0	473.14	47.00	2007
654.000	963.420	300.0	321.14	38.23	2007
647.880	939.890	150.0	626.59	51.40	2007

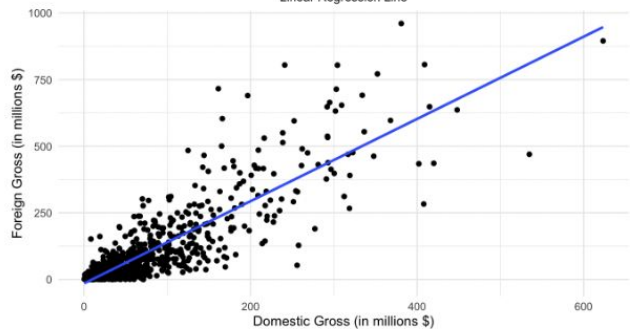
Analysis

- Regression analysis on what variables contribute to high movie profitability
- Potentially do cluster analysis with the movie genre or lead studio variables
- Create a model to predict how successful a movie will be

Beginning Exploratory Data Analysis

Relationship Between Domestic and Foreign Gross

Linear Regression Line

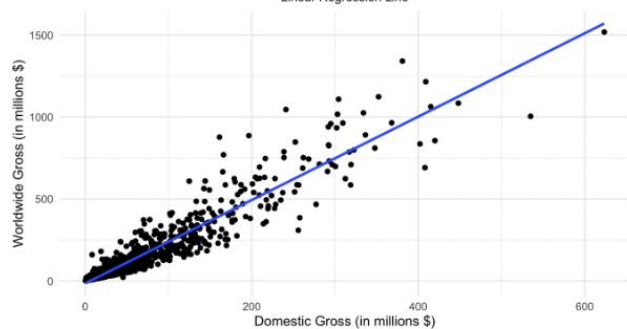


Data source: Movies dataset

Adjusted R-squared: 0.7212

Relationship Between Domestic Gross and Worldwide Gross

Linear Regression Line

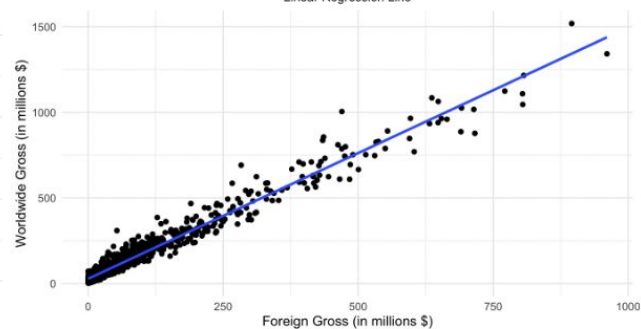


Data source: Movies dataset

Adjusted R-squared: 0.8753

Relationship Between Foreign Gross and Worldwide Gross

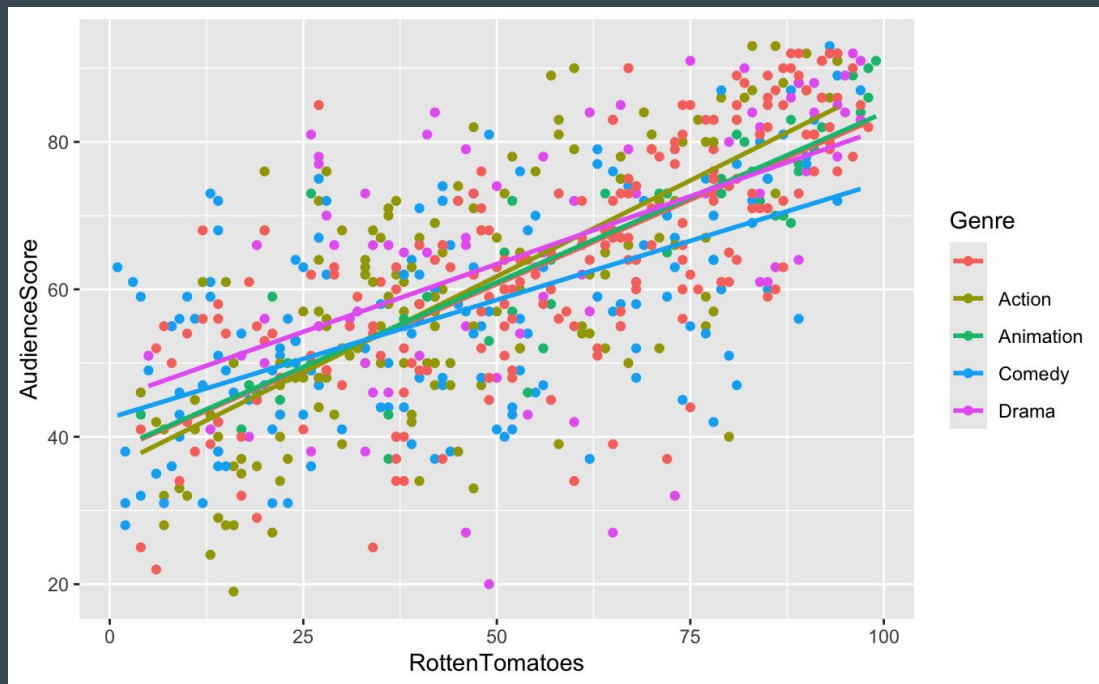
Linear Regression Line



Data source: Movies dataset

Adjusted R-squared: 0.9622

Beginning Exploratory Data Analysis



Some simple clustering (need to be improved)